

SQL Code

```
1  CREATE DATABASE ecommerce_analysis;
2  USE ecommerce_analysis;
3
4  CREATE TABLE customers (
5      customer_id INT PRIMARY KEY,
6      customer_name VARCHAR(100),
7      city VARCHAR(50)
8  );
9
10 CREATE TABLE products (
11     product_id INT PRIMARY KEY,
12     product_name VARCHAR(100),
13     category VARCHAR(50),
14     price DECIMAL(10,2)
15 );
16
17 CREATE TABLE orders (
18     order_id INT PRIMARY KEY,
19     customer_id INT,
20     product_id INT,
21     quantity INT,
22     order_date DATE,
23     FOREIGN KEY (customer_id) REFERENCES customers(customer_id),
24     FOREIGN KEY (product_id) REFERENCES products(product_id)
25 );
26
27 INSERT INTO customers VALUES
28 (11, 'Ananya', 'Hyderabad'),
29 (12, 'Vikram', 'Pune'),
30 (13, 'Meghana', 'Bangalore'),
31 (14, 'Rohit', 'Chennai'),
32 (15, 'Divya', 'Mumbai'),
33 (16, 'Karthik', 'Delhi'),
34 (17, 'Sneha', 'Hyderabad'),
35 (18, 'Aditya', 'Pune');
36
37 INSERT INTO products VALUES
38 (201, 'Tablet', 'Electronics', 28000.00),
39 (202, 'Wireless Mouse', 'Electronics', 1200.00),
40 (203, 'Bluetooth Speaker', 'Electronics', 3500.00),
41 (204, 'Office Chair', 'Furniture', 8500.00),
42 (205, 'Desk Lamp', 'Furniture', 1800.00),
43 (206, 'Running Shoes', 'Fashion', 4200.00),
44 (207, 'Hoodie', 'Fashion', 2500.00),
45 (208, 'Travel Bag', 'Accessories', 3000.00),
46 (209, 'Water Bottle', 'Accessories', 900.00);
47
48 INSERT INTO orders VALUES
49 (2001, 11, 201, 1, '2026-10-01'),
50 (2002, 12, 202, 2, '2026-10-02'),
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51 (2003, 13, 206, 1, '2026-10-03'),
52 (2004, 11, 203, 2, '2026-10-04'),
53 (2005, 14, 204, 1, '2026-10-05'),
54 (2006, 15, 207, 2, '2026-10-06'),
55 (2007, 16, 208, 1, '2026-10-07'),
56 (2008, 17, 205, 3, '2026-10-08'),
57 (2009, 18, 209, 4, '2026-10-09'),
58 (2010, 12, 203, 1, '2026-10-10'),
59 (2011, 13, 208, 2, '2026-10-11'),
60 (2012, 15, 206, 1, '2026-10-12');
61 select * from customers;
62 select * from products;
63 select * from orders;
64
65 SELECT *
66 FROM customers
67 WHERE city = 'Hyderabad';
68
69 SELECT
70     product_name,
71     category,
72     price
73 FROM products
74 ORDER BY price DESC;
75
76 SELECT
77     category,
78     COUNT(*) AS product_count
79 FROM products
80 GROUP BY category;
81
82 SELECT
83     SUM(p.price * o.quantity) AS total_revenue
84 FROM orders o
85 INNER JOIN products p
86     ON o.product_id = p.product_id;
87
88 SELECT
89     AVG(price) AS average_product_price
90 FROM products;
91
92 SELECT
93     p.category,
94     SUM(p.price * o.quantity) AS category_revenue
95 FROM orders o
96 INNER JOIN products p
97     ON o.product_id = p.product_id
98 GROUP BY p.category
99 ORDER BY category_revenue DESC;
100
101 SELECT
102     o.order_id,
103     p.product_name,
104     p.category,
```

```
105     o.quantity,
106     p.price
107 FROM orders o
108 INNER JOIN products p
109     ON o.product_id = p.product_id;
110
111 SELECT
112     c.customer_id,
113     c.customer_name,
114     c.city,
115     o.order_id,
116     o.order_date
117 FROM customers c
118 LEFT JOIN orders o
119     ON c.customer_id = o.customer_id;
120
121 SELECT
122     o.order_id,
123     p.product_name,
124     p.category,
125     p.price
126 FROM orders o
127 RIGHT JOIN products p
128     ON o.product_id = p.product_id;
129
130 SELECT
131     product_name,
132     category,
133     price
134 FROM products
135 WHERE price > (
136     SELECT AVG(price)
137     FROM products
138 );
139
140 CREATE VIEW revenue_analysis AS
141 SELECT
142     o.order_id,
143     c.customer_name,
144     p.product_name,
145     p.category,
146     o.quantity,
147     p.price,
148     (p.price * o.quantity) AS revenue
149 FROM orders o
150 INNER JOIN customers c
151     ON o.customer_id = c.customer_id
152 INNER JOIN products p
153     ON o.product_id = p.product_id;
154
155 SELECT *
156 FROM revenue_analysis;
157
158 CREATE INDEX idx_orders_customer_id
```

```
159 ON orders(customer_id);  
160  
161 SHOW INDEX FROM orders;  
162  
163
```